

Asset Centrality in Numerai

2023-04-08 / Numerai Meetup / [@yoshiso44](#)

Self Introduction

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Job:

botter / marker taker

Interests:

Financial Time Series Deep Learning /

Deep Learning Generalization on Noisy Data

Agenda

- What is Crowded Trades?
- What is Asset Centrality?
- Application on Numerai Data
- Future Works

What is Crowded Trades?

- Behavior of large capital inflow into similar context, like industry, sectors, and factor exposures, without having significant fundamental changes.
- Those caused by investor's psychological shift, or new strategy deployment.

What is Asset Centrality?

- Asset Centrality is a way to statistically estimate crowded trades
- Higher for more volatile or more connected sectors/factors
- Can be calculated with PCA

$$AR = \frac{\sum_{i=1}^n \sigma_{Ei}^2}{\sum_{j=1}^N \sigma_{Aj}^2}$$

$$C_i = \frac{\sum_{j=1}^n \left(AR^j \cdot \frac{|EV_i^j|}{\sum_{k=1}^N |EV_k^j|} \right)}{\sum_{j=1}^n AR^j}$$

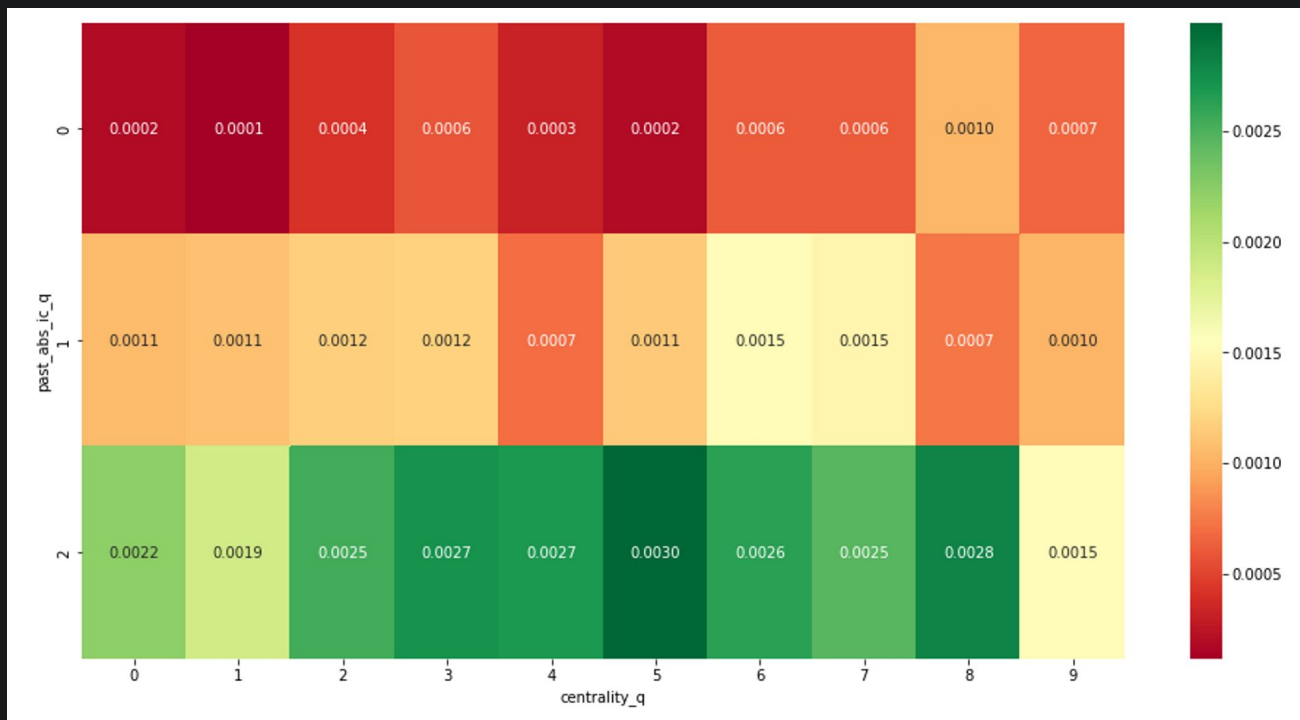
Asset Centrality in Numerai

- Features in Numerai can be treated as factors in traditional financial theory
- For simplicity, I'll treat each feature's IC as factor return and estimated Asset Centrality(AC) in Numerai's features

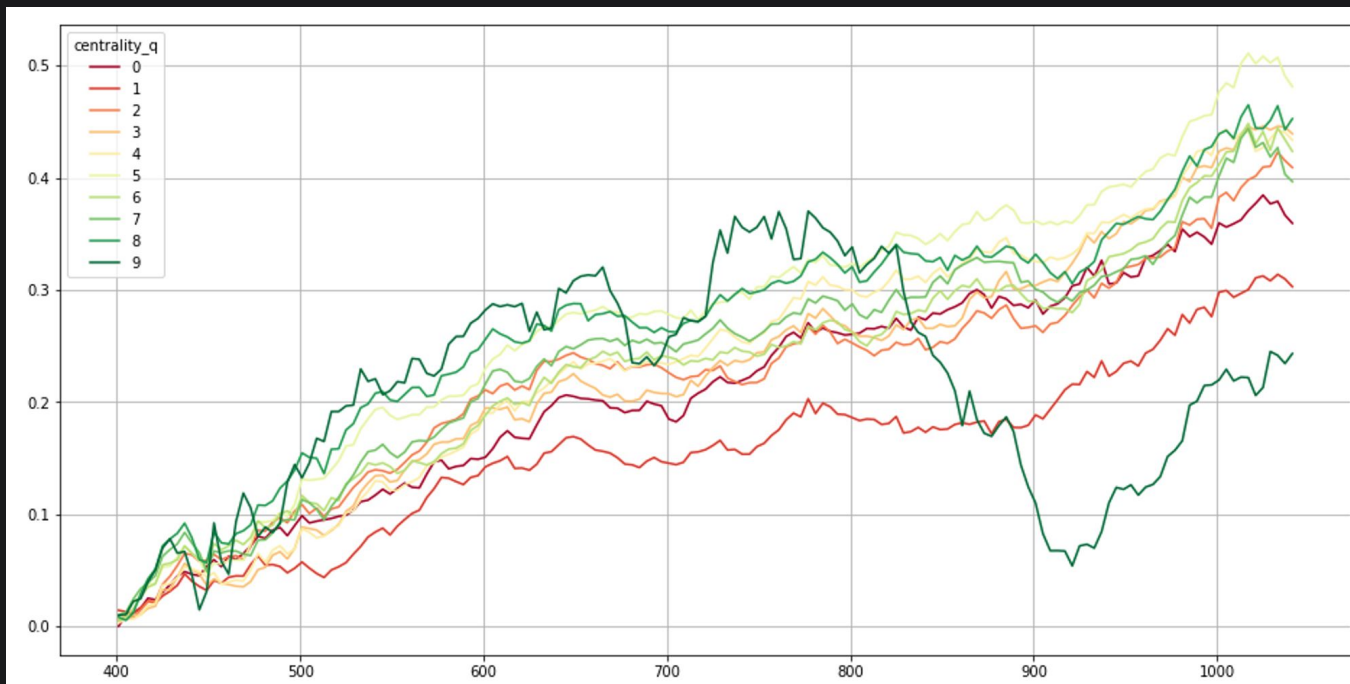
Experiment Setup

- Prepare IC data frame with calculating spearman correlation to target for each feature, each era.
- Estimate AC with using IC data frame.
- Check each feature's performance on absolute / relative AC quantiles.
- Check them especially on features performed well.

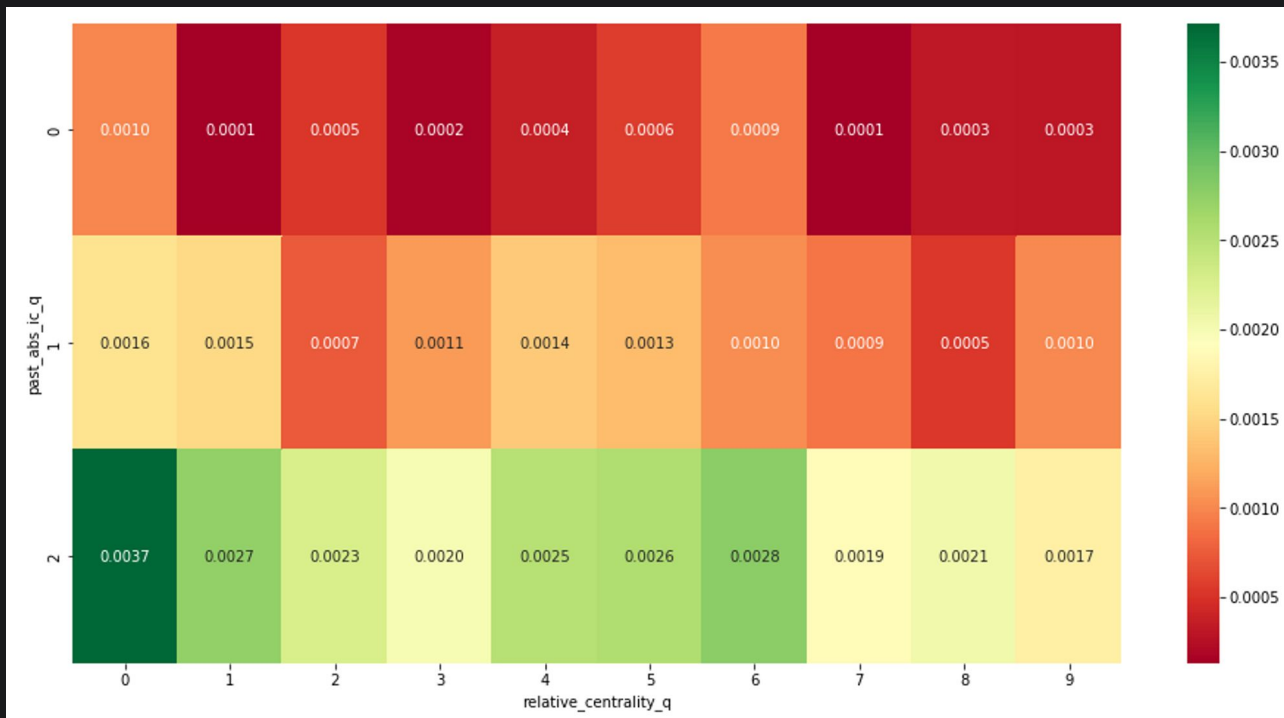
Performance on Absolute AC



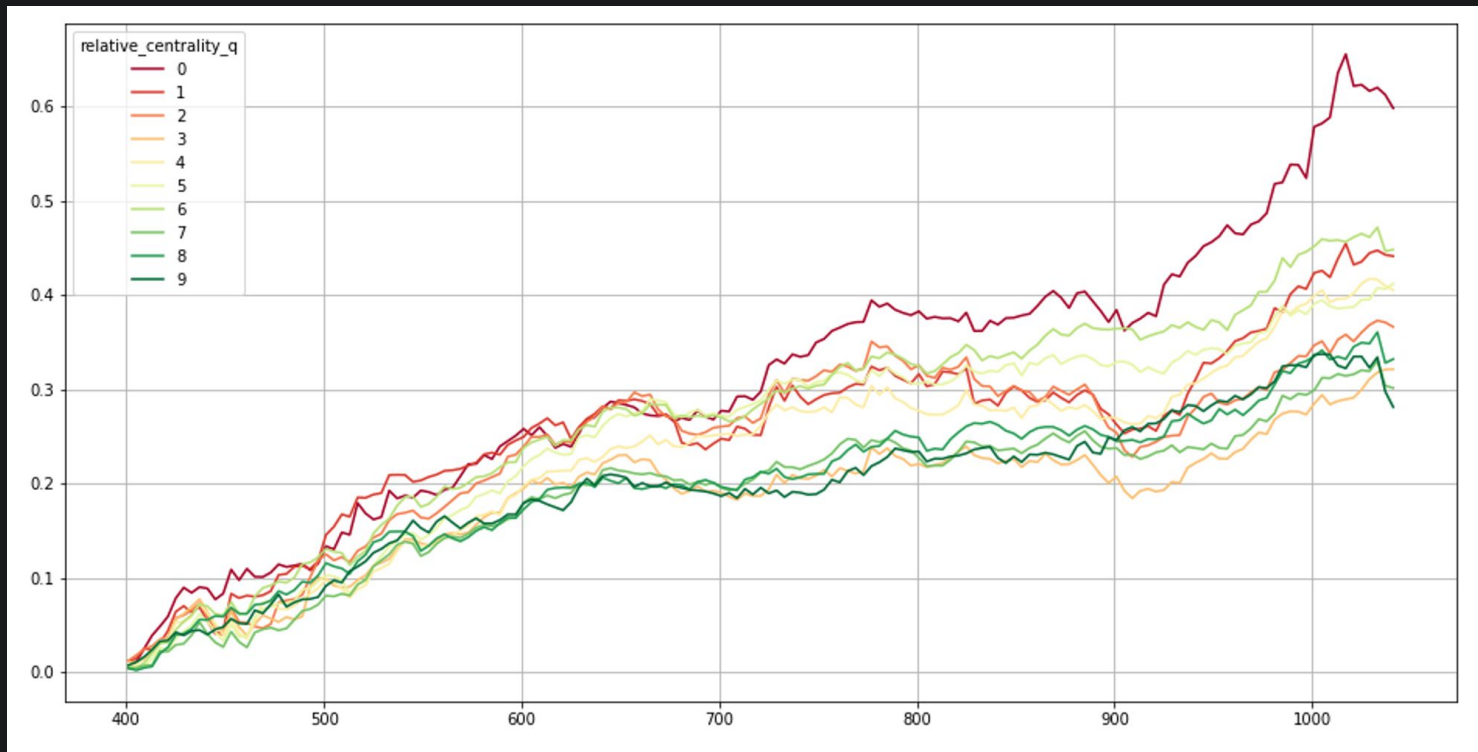
Performance on Absolute AC



Performance on Relative AC



Performance on Relative AC



Future Works

- Use it for modeling
- Exclude or neutralize highly centralized features.
- Dynamically control feature exposures to gain alpha via factor timing.

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